

Arm	Head	Loss	Features	Attention	Split	Status	R^2	Width / Pop_{RI}
<i>covariate</i>								
q_age_set	quantile	QuantileLoss	sex, age, setting	cross bidirectional	sequence	trained	0.718 ± 0.037	0.787 ± 0.122
334f7e21	quantile	QuantileLoss	sex	cross bidirectional	sequence	trained	0.717 ± 0.037	–
q_age	quantile	QuantileLoss	sex, age	cross bidirectional	sequence	trained	0.717 ± 0.037	0.682 ± 0.083
q_set	quantile	QuantileLoss	sex, setting	cross bidirectional	sequence	trained	0.719 ± 0.037	0.675 ± 0.082
q_co	quantile	QuantileLoss	sex, co-analytes	cross bidirectional	sequence	trained	0.717 ± 0.037	0.627 ± 0.082
q_age_co	quantile	QuantileLoss	sex, age, co-analytes	cross bidirectional	sequence	trained	0.717 ± 0.037	0.621 ± 0.080
q_set_co	quantile	QuantileLoss	sex, setting, co-analytes	cross bidirectional	sequence	trained	0.718 ± 0.037	0.612 ± 0.076
q_age_set_co	quantile	QuantileLoss	sex, age, setting, co-analytes	cross bidirectional	sequence	trained	0.719 ± 0.037	0.628 ± 0.081
q_co_q	quantile	QuantileLoss	sex, co-analytes + query	cross bidirectional	sequence	trained	0.718 ± 0.037	0.703 ± 0.095
<i>patient split</i>								
p_base	quantile	QuantileLoss	sex	cross bidirectional	patient	trained	0.718 ± 0.038	0.774 ± 0.113
p_co	quantile	QuantileLoss	sex, co-analytes	cross bidirectional	patient	trained	0.718 ± 0.038	0.670 ± 0.081
p_causal	quantile	QuantileLoss	sex	fully causal	patient	trained	0.717 ± 0.038	0.710 ± 0.083
p_full	quantile	QuantileLoss	sex, all past draws	fully causal, ≤ 128 tokens	patient	trained	0.715 ± 0.038	0.650 ± 0.079
<i>prior-anchored, multivariate</i>								
m_pa_k5	quantile	QuantilePriorLoss (an-chor, k=5)	sex, all past draws	fully causal, ≤ 128 tokens	patient	not trained	–	–
m_pa_k20	quantile	QuantilePriorLoss (an-chor, k=20)	sex, all past draws	fully causal, ≤ 128 tokens	patient	not trained	–	–
m_pa_tau	quantile	QuantilePriorLoss (an-chor, k=5, tau=365d)	sex, all past draws	fully causal, ≤ 128 tokens	patient	not trained	–	–
m_pf_k5	quantile	QuantilePriorLoss (floor, k=5)	sex, all past draws	fully causal, ≤ 128 tokens	patient	not trained	–	–
m_pg_k5	gate	QuantilePriorLoss (gate, k=5)	sex, all past draws	fully causal, ≤ 128 tokens	patient	not trained	–	–
m_pn_k5	nig	StudentTNLLLoss (nu0=5)	sex, all past draws	fully causal, ≤ 128 tokens	patient	not trained	–	–
m_gk_k5	gaussian	NORMALLoss (k=5, lambda=0.1, weighted by n)	sex, all past draws	fully causal, ≤ 128 tokens	patient	not trained	–	–
<i>prior-anchored</i>								
pa_k5	quantile	QuantilePriorLoss (an-chor, k=5)	sex, age, setting	cross bidirectional	sequence	started, no checkpoint	–	–
pa_k20	quantile	QuantilePriorLoss (an-chor, k=20)	sex, age, setting	cross bidirectional	sequence	partial, epoch 0	–	–
pa_tau	quantile	QuantilePriorLoss (an-chor, k=5, tau=365d)	sex, age, setting	cross bidirectional	sequence	partial, epoch 0	–	–
pf_k5	quantile	QuantilePriorLoss (floor, k=5)	sex, age, setting	cross bidirectional	sequence	partial, epoch 0	–	–
pg_k5	gate	QuantilePriorLoss (gate, k=5)	sex, age, setting	cross bidirectional	sequence	started, no checkpoint	–	–
pn_k5	nig	StudentTNLLLoss (nu0=5)	sex, age, setting	cross bidirectional	sequence	partial, epoch 24	–	–
gk_k5	gaussian	NORMALLoss (k=5, lambda=0.1, weighted by n)	sex, age, setting	cross bidirectional	sequence	trained	0.702 ± 0.038	0.969 ± 0.108

Table 1: Every NORMA arm trained since the covariate ablation began. All are NORMA2 at $d_{\text{model}} = 64$, 4 heads, 8 layers and 3 states, trained on EHRSHOT and MIMIC-IV together. Every arm conditions on the analyte identity and on age at the first draw; *Features* lists the per-draw covariates each arm uses, sex included. R^2 is the test R^2 on the development split and *Width / Pop_{RI}* the mean 95% interval width divided by the population reference interval width on normal-state queries; both are means over analytes weighted by the number of targets, pooled over EHRSHOT and MIMIC-IV, $\pm$ half a 95% bootstrap interval over analytes (2,000 resamples). They are comparable within a split group and only indicative across them, since the patient-split arms are scored on different rows. A width below 1 is narrower than the population interval. *Attention*: the decoder layers are called with memory equal to target, so the causal mask applies to self-attention only,